

raw_tool_call → ActionRequest → DecisionResponse → ExecutionResult → AuditEvent → ActionTrace

1

Raw Tool Call

INPUT

Unstructured output from the LLM / agent — the starting point of every guarded action.

user_goal the user's original intent

tool_call raw model tool name + args

normalize

2

ActionRequest

NORMALIZE

Canonical, validated action record that everything downstream keys off. **app/core/action_schema.py**

schema_version "0.1"

action_type e.g. file_read

target_system system being acted on

payload_summary safe summary

browser_element optional

rollback_available bool

content_context page/selection context for LLM decisions · created_at

run_id / action_id correlation ids

domain / source e.g. productivity / cli

target str or dict

payload excluded from JSON

risk_hint heuristic risk

confidence 0.0–1.0

guardrail decision

3

DecisionResponse

GUARDRAIL

Policy/risk verdict on the action before anything is executed. **app/core/action_schema.py**

decision ALLOW / BLOCK / NEED_APPROVAL / SANITIZE / ASK_USER

reasons human-readable rationale

sensitive_entities detected PII etc.

next_step execute / approval_queue / sanitize / ask_user

risk_level / risk_score LOW–CRITICAL / 0–1

triggered_policies policy ids hit

sanitized_payload when SANITIZE

latency_ms decision time

ALLOW

proceed to execution →

NEED_APPROVAL

pending, halt until approved

BLOCK

action rejected, never execute

SANITIZE

redact payload, then confirm

ASK_USER

clarification required, no execution

execute (permitted decisions only)

4

ExecutionResult

EXECUTE

Outcome produced by a concrete executor (browser, file, API...). **app/core/executor_schema.py**

executor which executor ran

result_summary short outcome text

error details on failure

status SUCCESS / FAILED / SKIPPED / BLOCKED / PENDING_APPROVAL /
SANITIZED / WAITING_USER

data structured result

latency_ms execution time

persist

5

AuditEvent

AUDIT

Immutable, queryable event stored for the audit trail. **app/core/audit_schema.py**

audit_id event id

request_json action request snapshot

execution_json result snapshot

error_type when failed

latency stage timings · created_at

run_id / action_id correlation ids

decision_json decision snapshot

execution_status final status

policy / detector_version "policy-0.1" / "detector-0.1"

assemble trace

6

ActionTrace

TRACE

Full end-to-end record of one action: every stage captured for replay/debugging. **app/core/audit_schema.py**

user_goal original intent

raw_tool_call stage 1 snapshot

decision stage 3 snapshot

audit stage 5 snapshot

latency stage timings · created_at

run_id / action_id correlation ids

action_request stage 2 snapshot

execution stage 4 snapshot

final_status outcome

PERSISTENCE

artifacts/audit/events.jsonl — audit events

artifacts/traces/actions.jsonl — action traces

READ APIS (APP/API/V1)

GET /runs · GET /runs/{id}/actions

GET /actions/{action_id} · GET /audits

GET /audits/latest · GET /approvals · GET /benchmark